Towns with an Asteroid

The places on Earth that minor planets are named after,
and the observatories that named them

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DOI: https://doi.org/10.5281/zenodo.24052725

2026

Abstract

Discoverers of asteroids propose their names, and many choose places. Using Wikidata's "named after" statements for 11,268 asteroids, I identify 2,206 namings of places with coordinates, on 2,193 asteroids (19.3% of all namings; 65.4% honour people) and link each place to the site where its asteroid was discovered. Towns, cities and villages make up 52.1% of the places. Japan has the most (421); per head, Finland (11.8 per million people), the Czech Republic (9.8) and Sweden (8.1) lead. The median distance from discovery site to named place is 508 km and 27.1% of places lie within 100 km. Naming geography follows the astronomer: only 3.6% of the places named from La Silla in Chile are Chilean, while Kleť Observatory named 98 places, 80.6% of them Czech and 50.0% within 100 km, and 92.5% of Japanese places were named from Japanese observatories. I describe the data, its incompleteness and the classification, and publish the linked tables and an interactive map.

Keywords: minor planets; asteroids; nomenclature; toponymy; commemoration; amateur astronomy; Wikidata; open data; mapping

1 Introduction

Once an asteroid's orbit is secure it receives a number, and its discoverer may propose a name, which the International Astronomical Union's Working Group Small Bodies Nomenclature considers and publishes (Minor Planet Center, 2026; WGSBN, 2026). The result is a large, informal archive of what astronomers chose to remember: colleagues and family, scientists and artists, and places. Many place names are personal. An observatory names its own town and the villages around it; an astronomer abroad names the town where they grew up.

This paper maps that naming geography with open data. Wikidata (Vrandečić and Krötzsch, 2014) records, for many asteroids, what they are named after, where they were discovered and by whom. Joining the two gives each named place a line to the observatory that found its asteroid. I ask which places have an asteroid, which countries have most, who named them, and how close to home.

2 Data

A single query of Wikidata through QLever (Bast and Buchhold, 2017) returned every item that is an asteroid (an instance of asteroid or of a subclass) with a "named after" (P138) statement, with its minor planet number, discoverer (P61), discovery site (P65) and its coordinates and country, discovery date, and the namesake's type, coordinates and country. The query returned 22,704

rows, one per discoverer and type, describing 11,268 asteroids and 11,418 asteroid and namesake pairs ("namings"); 133 asteroids have more than one namesake. Wikidata is released under CC0. Natural Earth (Natural Earth, 2026) gives modern countries and their population estimates.

Wikidata does not record the namesake of every named minor planet, and I have not compared these records with the Minor Planet Center's complete list of names. The data are therefore the subset of named minor planets that Wikidata links to a namesake, not all named minor planets, and its coverage may differ by country and period.

3 Methods

3.1 Places and kinds

A namesake is a person if Wikidata says it is human (Q5); otherwise it is a place if it has coordinates and is not mythical or fictional. Namesakes that have coordinates but are not places are also "other": any whose Wikidata types include a language, a taxon, a ship, a battle or other event, a television programme, a people or ethnic group, a dynasty or a troupe; bodies whose types are all organisational (an agency, society, ministry, survey or league with no site of its own); and two cases found by hand (a star party and a government department typed as an observatory and a region). I read the whole resulting list of places by eye. Everything else (gods, works, organisations without coordinates) is "other" too. Places were sorted into kinds by keyword rules applied, in order, to all of their Wikidata types: rivers, lakes and seas; towns, cities and villages; countries (exact types such as "sovereign state" or "historical country"); observatories, universities and other institutions; mountains, islands and other natural features; and regions, provinces and counties. A small remainder is "other places".

3.2 Countries, distances and home

Each place and each discovery site is given the modern country Wikidata records for it (P17) when that names exactly one modern country and the point lies within about 30 km of that country's Natural Earth outline, which settles border towns such as Passau, Konstanz, Gorizia and Tornio; otherwise (no country, several, or only a historical state) it is placed in a Natural Earth country by its coordinates. Continents are assigned to no country. Crimea, where Natural Earth's default view places it in Russia, is counted as Ukraine. Per head rates use Natural Earth's population estimates (mostly for 2019) and are reported for countries of at least a million people with five or more place namings. Populations are Natural Earth country figures, so Hong Kong and other territories count separately. Distances are great circle distances between the discovery site and the place's single point, so they are rough for regions, rivers and countries. A place is "home" if it is in the same country as the discovery site.

4 Results

4.1 What is named

Of the 11,418 namings, 7,463 (65.4%) honour people, 2,206 (19.3%) places with coordinates and 1,749 (15.3%) other things (Figure 2). The 2,206 place namings cover 2,175 distinct places: 1,150

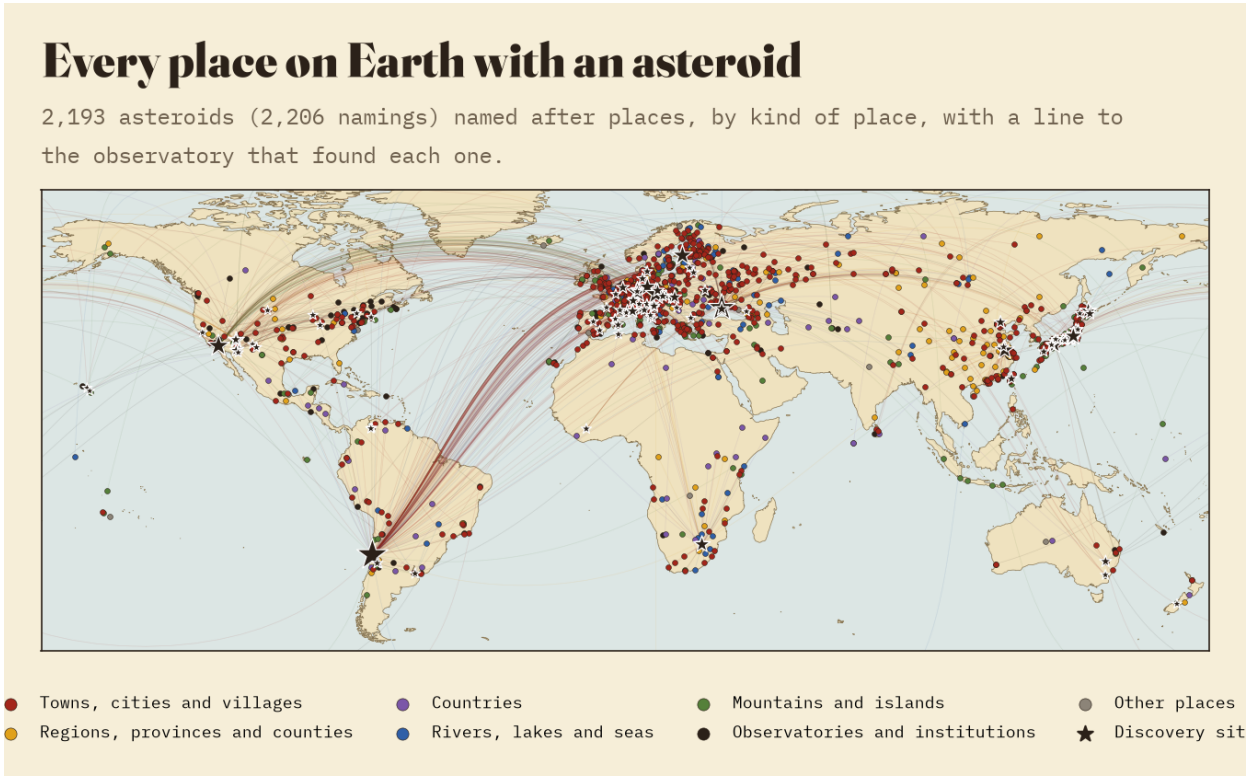


Figure 1: Every place with an asteroid in Wikidata, by kind, with great circle lines to the discovery sites.

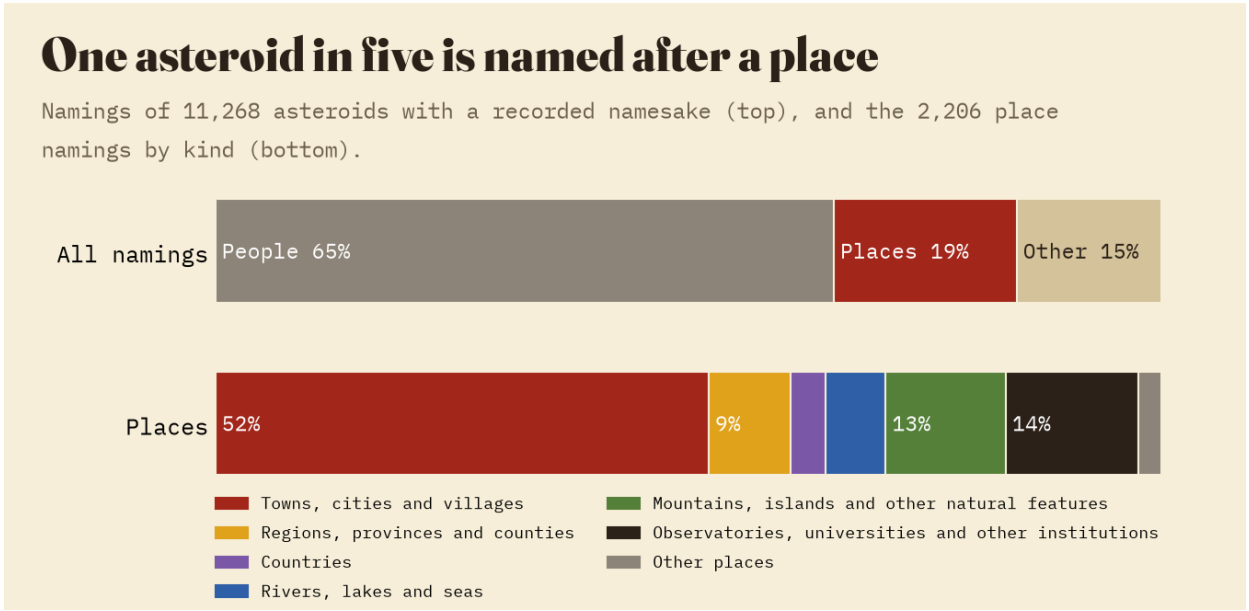


Figure 2: People, places and other namesakes (top) and the kinds of place (bottom).

namings of towns, cities and villages, 309 of institutions, 280 of natural features, 191 of regions, 140 of rivers and lakes and 82 of countries. Figure 1 maps them all.

4.2 Countries

Places in 111 modern countries have an asteroid. Japan leads with 421 (19.1%), followed by Germany (153), Russia (137), Italy (135) and the United States (130) (Table 1, Figure 3). Per head, Finland, the Czech Republic and Sweden lead, each with long traditions of asteroid discovery.

Table 1: The fifteen countries with the most place namings.

Country	Places	Share (%)	Per million
Japan	421	19.1	3.33
Germany	153	6.9	1.84
Russia	137	6.2	0.95
Italy	135	6.1	2.24
United States	130	5.9	0.40
China	120	5.4	0.09
France	106	4.8	1.58
Czech Republic	104	4.7	9.75
Sweden	83	3.8	8.07
Finland	65	3.0	11.77
Ukraine	64	2.9	1.44
Spain	49	2.2	1.04
Belgium	43	1.9	3.74
Switzerland	32	1.4	3.73
Hungary	32	1.4	3.28

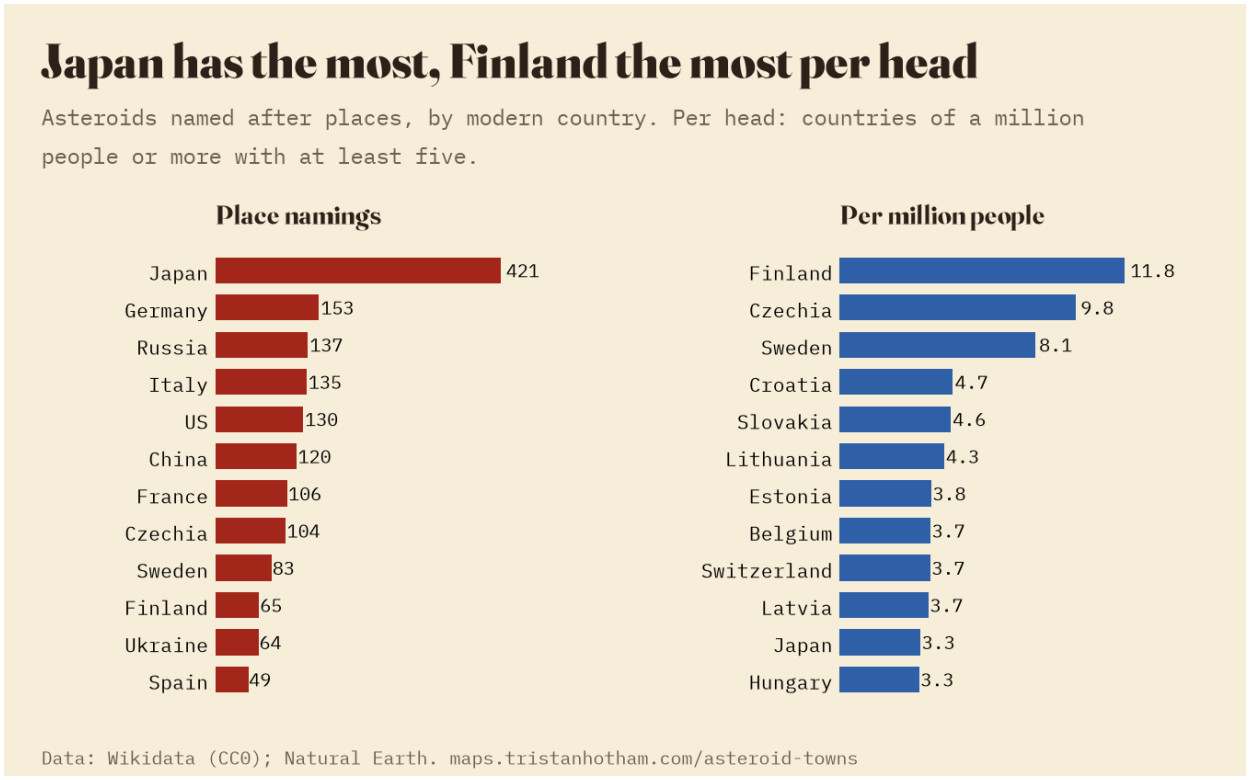


Figure 3: Place namings by country, in total and per million people.

Table 2: The fifteen discovery sites with the most place namings: share in the site's own country, median distance and share within 100 km.

Site	Country	Places	Own (%)	Median km	<100 km (%)
La Silla Observatory	Chile	224	3.6	11,631	1.8
Crimean Astrophysical Observatory	Ukraine (Crimea)	164	25.6	1,251	9.8
Palomar Observatory	United States	99	21.2	9,177	1.0
Kleť Observatory	Czech Republic	98	80.6	101	50.0
Heidelberg-Königstuhl State Observatory	Germany	86	27.9	795	14.0
Iso-Heikkilä Observatory	Finland	81	79.0	364	13.6
Purple Mountain Observatory	China	72	98.6	993	5.6
Karl Schwarzschild Observatory	Germany	63	74.6	285	25.4
Ōizumi Observatory	Japan	63	96.8	120	41.3
Kitami Observatory	Japan	51	96.1	245	27.5
Union Observatory	South Africa	49	34.7	1,471	2.0
Geisei observatory	Japan	48	85.4	117	45.8
Anderson Mesa Station	United States	44	22.7	9,089	4.5
Piszkéstető Station	Hungary	30	93.3	174	36.7
Simeiz Observatory	Ukraine (Crimea)	28	32.1	976	21.4

4.3 Who named them

The 237 discovery sites are led by La Silla (224 place namings), the Crimean Astrophysical Observatory (164) and Palomar (99) (Table 2, Figure 4). Large observatories host visiting astronomers, and their names travel with them: 3.6% of La Silla's places are in Chile. The Belgian astronomer Eric Elst, the most prolific namer of places in the data (167), worked at La Silla and named 36 places in France and 16 in Belgium; the Uppsala survey at La Silla named Swedish places, so that none of Sweden's 83 places was named from Sweden. Finland is the opposite: all 65 of its places were named from Finnish sites, most by Yrjö Väisälä's observatory at Turku. Purple Mountain Observatory named Chinese places almost exclusively (98.6%).

4.4 Close to home

The distribution of distances is wide (Figure 5). The median is 508 km; 27.1% of places lie within 100 km of the discovery site, and 204 within 10 km, many of them observatories named after themselves (26 of the 82 asteroids named after observatories or telescopes were found there). 58.5% of places are in the discovery site's country. The most local namers are small observatories with a resident team (Figure 6): Kleť in South Bohemia (50.0% within 100 km), and Japanese sites such as Nanyo (68.2%) and Geisei (45.8%). Of the places in Japan, 92.5% were named from Japanese sites and 46.1% from within 100 km; for the Czech Republic the figures are 90.4% and 54.8%, and for Italy 49.2% and 37.7%.

4.5 The United Kingdom

26 asteroids are named after places in the United Kingdom (16 towns and cities). Nearly all were discovered abroad. The exceptions are an amateur observatory at Church Stretton, Shropshire, which found the asteroids named after Church Stretton and Shropshire (Salopia), and one at Stakenbridge, which found Old Joe, named after the University of Birmingham's clock tower (Table 3).

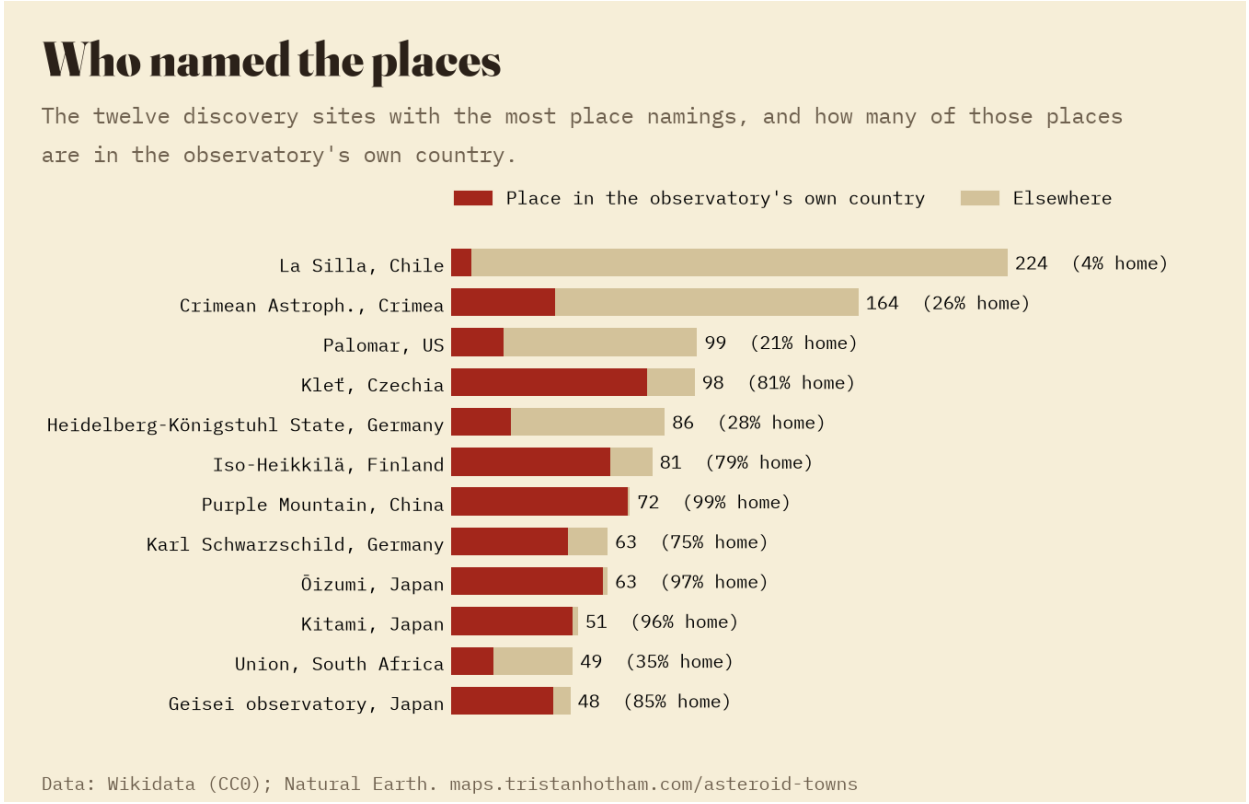


Figure 4: The twelve discovery sites with the most place namings, and the share in their own country.

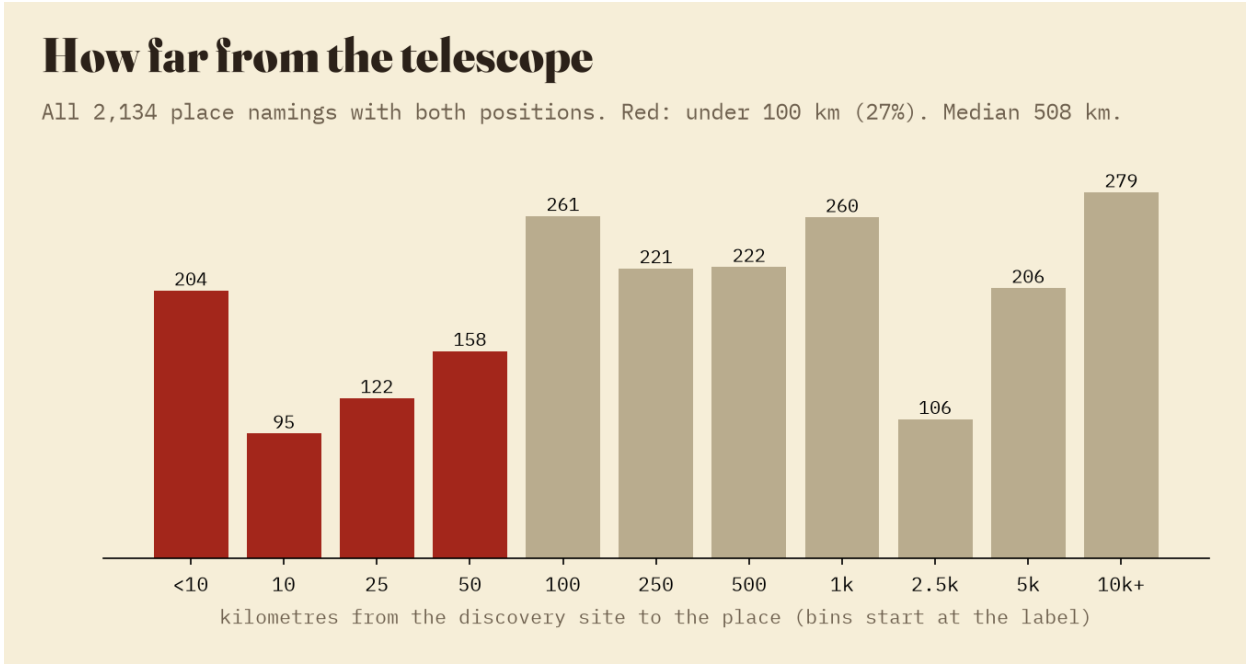


Figure 5: Distance from discovery site to named place.

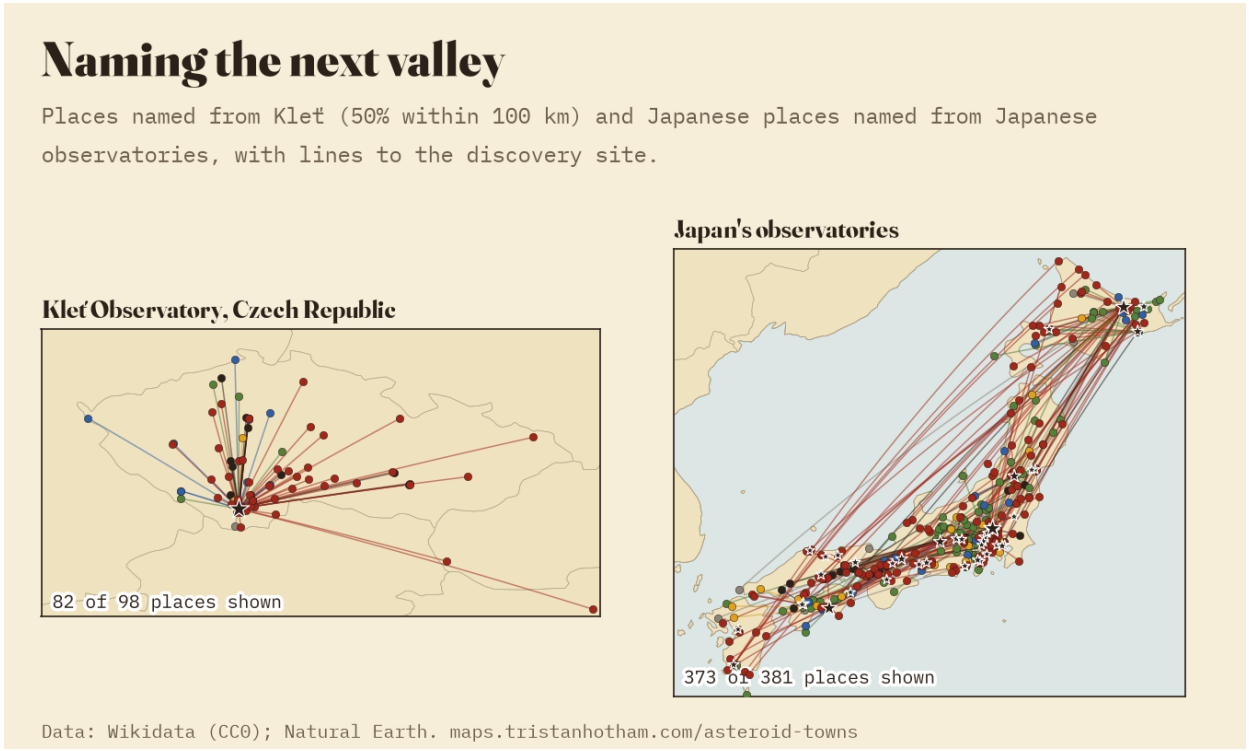


Figure 6: Local naming: places named from Kleť Observatory, and places in Japan named from Japanese observatories.

Table 3: Asteroids named after places in the United Kingdom.

Asteroid	Place	Discovery site	Year	km
758 Mancunia	Manchester	Union Observatory	1912	9,328
1071 Brita	Great Britain	Simeiz Observatory	1924	2,811
1244 Deira	Kingdom of Deira	Union Observatory	1932	9,329
2531 Cambridge	Cambridge	Anderson Mesa Station	1980	8,270
2830 Greenwich	Royal Observatory	Anderson Mesa Station	1980	8,313
3009 Coventry	Coventry	Crimean Astrophysical Observatory	1973	2,719
3858 Dorchester	Dorchester	Brorfelde Observatory	1986	1,084
4206 Verulamium	Verulamium	La Silla Observatory	1986	11,306
5677 Aberdonia	University of Aberdeen	Lowell Observatory	1987	7,824
5805 Glasgow	Glasgow	Lowell Observatory	1985	7,798
6939 Lestone	Leighton Buzzard	Mount Wilson Observatory	1952	8,672
7603 Salopia	Shropshire	Church Stretton	1995	11
8283 Edinburgh	Edinburgh	Siding Spring Observatory	1991	16,524
8380 Tooting	Tooting	Palomar Observatory	1992	8,751
8837 London	London	La Silla Observatory	1989	11,305
8849 Brighton	Brighton	La Silla Observatory	1990	11,263
9325 Stonehenge	Stonehenge	La Silla Observatory	1989	11,186
9767 Midsomer Norton	Midsomer Norton	Siding Spring Observatory	1992	16,796
10501 Ardmacha	Armagh	Palomar Observatory	1987	8,209
10502 Armaghobs	Armagh Observatory	Palomar Observatory	1987	8,209
10515 Old Joe	Joseph Chamberlain Memorial Clock Tower	Stakenbridge observatory	1989	16
11626 Church Stretton	Church Stretton	Church Stretton	1996	0
28220 York	York	Kleť Observatory	1998	1,203
69423 Openuni	The Open University	Astronomical Observatory of Farra d'Isonzo	1996	1,238
100604 Lundy	Lundy	Royal Observatory of Belgium	1997	633
129092 Snowdonia	Snowdon	Haleakalā Observatory	2004	11,384

5 Limitations

First, coverage: Wikidata records the namesake of only part of the named minor planets, and the share may vary by period, country and discoverer. Second, the observatory is the place of discovery, not certainly the place of naming, and names are sometimes proposed by people other than the discoverer. Third, kinds come from keyword rules on Wikidata's types and are approximate. Fourth, regions, rivers and countries are single points, so their distances are rough. Fifth, modern borders are applied throughout, and Crimea is assigned to Ukraine.

6 Conclusion

About one asteroid in five with a recorded namesake honours a place. Japan's dense network of amateur and municipal observatories has put more of its towns in the sky than any other country, and small European observatories such as Kleť have named their own valleys. Large international observatories spread names around the world, carried by visiting astronomers. Linking open data on asteroids and places makes this naming geography visible and checkable.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/asteroid-towns/. Data and code are at https://github.com/Tristan-Hotham/asteroid-towns and archived with this paper at https://doi.org/10.5281/zenodo.24052725. Derived data and text are released under CC BY 4.0 (the Wikidata source is CC0); code under the MIT licence.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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